

Machine Learning for teens

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K nearest

- ▶ KNN is supervised learning algorithm
- ▶ In KNN, we find the nearest point, and our point belongs to the same class as that nearest point
- ▶ KNN in Two Simple Steps
 - ▶ Find the nearest data point(s)
 - ▶ Assign the class of the nearest neighbor(s)

Kmean clustering

- ▶ K-Means
 - ▶ Unsupervised learning algorithm
 - ▶ groups data into **K clusters** based on similarity.
- ▶ We want to find the best value for k. To do this, we use the elbow method, where we plot the Sum of Squared Errors (SSE) for different values of k. We then look at the SSE vs. k plot and choose the value of k where an 'elbow' appears in the curve. This point represents the optimal number of clusters."

Kmean clustering- Elbow method

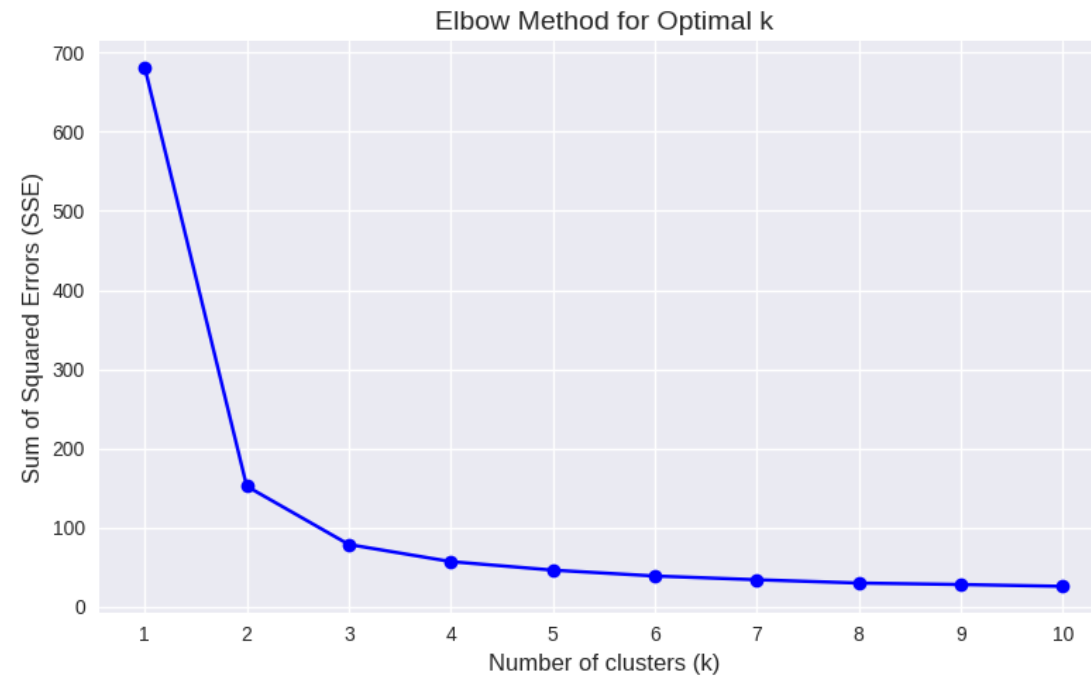
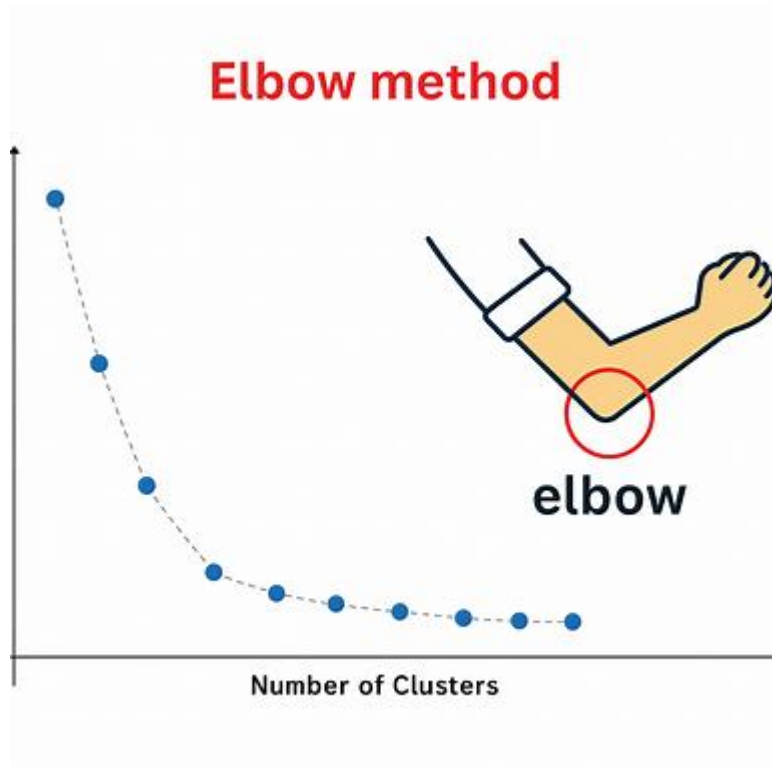
- SSE for each cluster

$$SSE_i = \sum_{x \in C_i} (x - \mu_i)^2$$

- Sum of SSE

$$SSE = SSE_1 + SSE_2 + \dots + SSE_k$$

Kmean clustering- Elbow method



assignment

- ▶ Load Mall_Customers.csv
- ▶ Apply k-mean clustering and find optimal k value
- ▶ Then use KNN to do prediction
- ▶ Finally, perform hyperparameter tuning for all models using GridSearchCV.